Kansas Frontier Matrix (KFM)

Comprehensive Data Source Integration Blueprint

Extensive, build-ready engineering plan to integrate the complete KFM Data Sources inventory into the KFM system design.

Boss mode summary (non-negotiables):
1) **Trust membrane**: UI and external clients never access databases directly - all access is via the governed API + policy boundary.
2) **Fail-closed policy** on every request (data, Story Nodes, AI).
3) **Dataset promotion gates**: Raw -> Work -> Processed; promotion requires checksums + STAC/DCAT/PROV catalogs.
4) **Focus Mode must cite or abstain**; every answer produces an audit reference.

Version: 1.0 **Date:** 2026-02-12
Primary input: *KFM Data Sources.pdf* (governed internal reference).
Secondary inputs: *KFM Next-Gen Blueprint & Primary Guide v1.1 (draft)*; *KFM Integration Engineering Brief v0.3*.

Status: Engineering blueprint (suitable for architectural review, interface contracts, and CI gates).

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1. Purpose, scope, and definitions

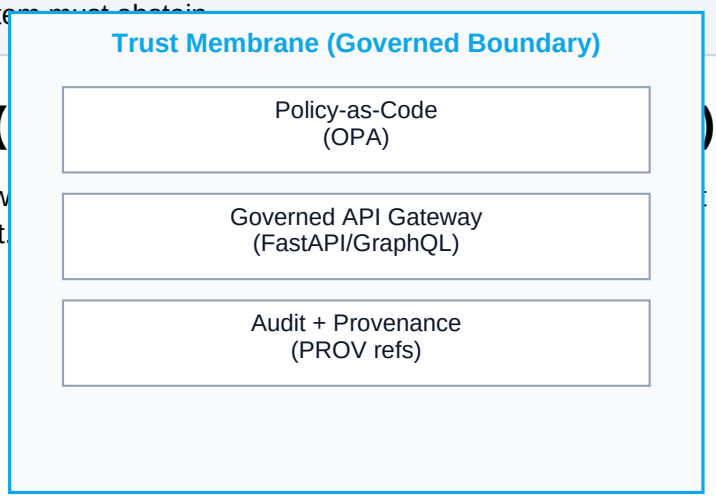
This document converts the KFM Data Sources inventory into an engineered integration plan: how each source is acquired, validated, versioned, cataloged, stored, indexed, and exposed through the KFM trust membrane to the Web UI and Focus Mode.

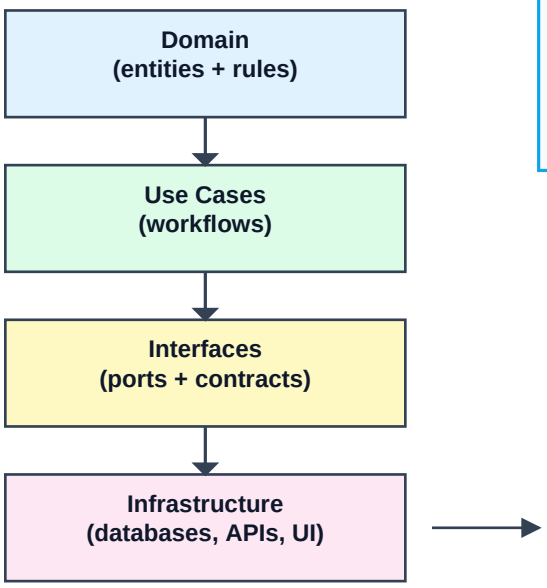
The emphasis is build-readiness: boundaries, contracts, validation gates, and acceptance criteria are named explicitly so the system cannot regress into ad-hoc pipelines or uncited narratives.

Evidence-first contract: Every user-visible claim must link to a dataset version and the exact record(s) used. If evidence cannot be resolved, the syste

2. Architecture blueprint (

KFM is designed as a cleanly layered system w
membrane via policy-as-code on every request.





2.1 Layer responsibilities (what goes where)

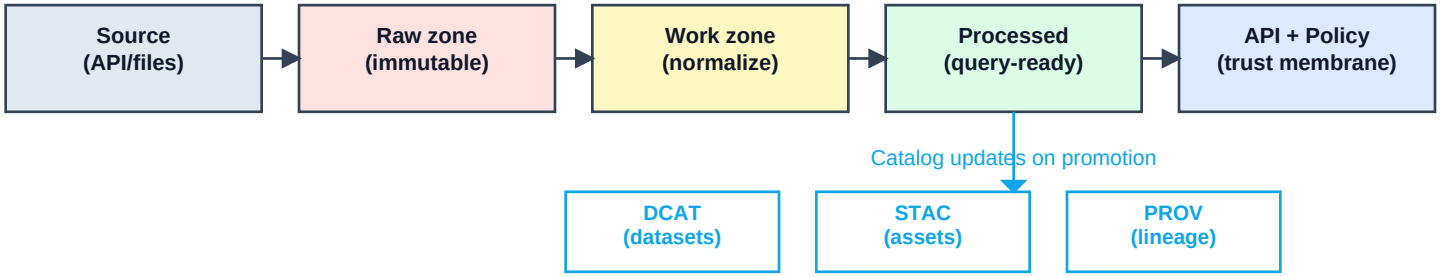
- **Domain**: entities (Place, Event, Observation, Artifact, Dataset, DatasetVersion) and invariants (time validity, provenance requirements).
- **Use cases**: ingest datasets, promote versions, resolve evidence, build story nodes, run entity resolution workflows.
- **Interfaces**: repository contracts, adapter contracts, policy interface, catalog writers, external service ports.
- **Infrastructure**: PostGIS, object storage, graph/search indices, FastAPI/GraphQL, React/MapLibre UI, CI/CD runtime.

2.2 Trust membrane mechanics

All reads and writes crossing the trust membrane pass through authentication, policy evaluation, query shaping/redaction, and audit/provenance logging. This is not optional - it is a build invariant enforced by tests.

3. Data zones, catalogs, and promotion gates

Every dataset flows through three zones. Promotion is only allowed when machine-checkable catalogs and validations succeed.



Zone	Purpose	Allowed operations	Required artifacts for promotion
Raw	Immutable capture of source-of-truth data.	Append-only writes; no transforms.	Checksums + raw manifest; source license captured.
Work	Repeatable transformations and QA staging.	Derivation, normalization, enrichment.	PROV activity; intermediate QA reports.
Processed	Query-ready datasets exposed via API.	Read-optimized storage + indexing.	DCAT dataset; STAC collection/items (if spatial); PROV chain; signed policy labels.

3.1 Catalog standards (machine-readable)

- **DCAT**: dataset-level metadata (publisher, license, temporal/spatial coverage, update frequency, contacts).
- **STAC**: geospatial assets (rasters/vectors) for map/timeline rendering; one Collection per product, Items per time/area unit.
- **PROV**: transformation lineage; every derivative artifact must reference its inputs and processing activity.

3.2 Promotion gate checklist (CI-enforced)

- All raw assets checksummed and addressable by content hash.
- Schema validation passes; QA report stored as an artifact with a stable ID.
- Policy labels attached (public, restricted, sensitive-location, etc.).

- Catalog writers succeed (DCAT/STAC/PROV are well-formed and link-check clean).
- Contract tests for API queries that depend on the dataset pass.

4. Canonical data model and ID strategy

KFM integrates heterogeneous sources by mapping them into a small set of canonical concepts. This enables consistent joins, governance, and evidence citations.

Entity	What it represents	Key fields (minimum)	Backends
Dataset / DatasetVersion	A governed source and a specific ingested version.	dataset_id, version_id, license, checksum_manifest, time_range, spatial_extent	Catalog + object store
Place (GeoID)	A time-valid geographic unit (point/line/polygon).	geoid, geometry, valid_from/to, place_type, source	PostGIS + graph
Observation	A measured or reported value tied to time and place.	obs_id, time, place_geoid, value, unit, method, source_record_id	PostGIS + analytics
Event	A time-bounded occurrence (flood, tornado, patent issuance).	event_id, time_range, footprint, event_type, participants, source_record_ids	Graph + PostGIS
Artifact	A document/image/map/newspaper page with metadata.	artifact_id, uri, mime, derived_from, ocr_text, citations	Object store + search/vector

4.1 Identifier rules (deterministic + stable)

- Dataset IDs are stable (publisher + product + scope). DatasetVersion IDs are content-addressed (hash of raw manifest + metadata).
- All user-visible evidence citations reference DatasetVersion + source_record_id(s).
- Places are time-valid: the same named unit (e.g., a county) may have multiple geometries across vintages.
- Every transformation produces new identifiers; never overwrite prior versions.

4.2 Time model

Use ISO-8601 timestamps with explicit time zones (or UTC) and represent uncertain time ranges as [start, end] intervals. For historical sources with fuzzy dates (e.g., newspapers), store both the publication date and the event date claim (if extracted), each with its own provenance.

5. Ingestion & processing framework

KFM ingestion is connector-driven. Each data source is implemented as an adapter that conforms to a common contract: discover -> acquire -> normalize -> validate -> publish. This enables consistent governance and reduces one-off code.

5.1 Connector contract (interface)

```
interface DataSourceConnector {
  discover(ctx): Capabilities
  acquire(ctx, plan): RawManifest
  transform(ctx, manifest): WorkArtifacts
  validate(ctx, work): ValidationReport
  publish(ctx, work, report): DatasetVersionRef
}
```

5.2 Orchestration and backfills

- Schedule connectors by cadence (real-time, daily, weekly, annual, static).
- All jobs must be idempotent: re-running a job never mutates a published DatasetVersion.
- Backfills (historic ranges) are treated as explicit runs with their own audit trail.
- Operational metadata: run start/end, rows read/written, error counts, latency, freshness.

5.3 Cross-source alignment (joins that make stories possible)

- Geography: normalize to GeoIDs and maintain boundary vintages + crosswalks.
- Time: normalize to a shared time model; store both reported and derived timestamps.
- Entity resolution: match people/orgs/places cautiously and record confidence; never merge silently.
- Events: create event nodes and link supporting observations/artifacts as evidence.

6. Governed API blueprint (data + evidence + Focus Mode)

All downstream consumers use the governed API. The API is responsible for policy evaluation, evidence resolution, query shaping/redaction, and audit logging.

Endpoint family	Purpose	Key outputs
/datasets, /catalog	Discover datasets and versions; expose DCAT/STAC entries.	Dataset list, version refs, catalog URLs
/map/tiles, /features	Map rendering primitives from Processed zone.	Vector features, raster tiles, footprints
/timeseries, /observations	Time series retrieval and aggregation.	Observations (with provenance)
/evidence/resolve	Translate a claim into evidence packages.	Record IDs, snippets, media, provenance chain
/focus/query	Focus Mode orchestration (questions -> evidence -> answer).	Answer + citations or abstain + audit ref

6.1 Policy-as-code (OPA) integration points

- Dataset/Version access (public vs restricted).
- Field-level redaction (e.g., ownership names, sensitive coordinates).
- Aggregation thresholds (health + crime to prevent reidentification).
- Derivative publication rules (public summaries vs restricted raw).
- Audit requirements: every response carries an audit reference and evidence bundle hash.

7. Data source register (from KFM Data Sources)

This section translates the KFM Data Sources inventory into a system-ready registry. Each source is expanded into a buildable implementation profile in Appendix A.

Domain	Data source	Access	Cadence	License	Sensitivity
Biodiversity	GBIF - Global Biodiversity Information Facility	Open API + bulk downloads (DwC-A/CSV)	Continuous / as published	Open data (varies by record; typically CC0/CC BY)	Low
Biodiversity	iNaturalist Observations	Open API + bulk exports	Continuous	Varies (observation/media licenses)	Medium (precise locations for sensitive species)
Climate	NOAA NCEI - Historical Climate Data (Climate Data Online)	API + bulk	Daily/monthly updates	US Gov public domain	Low
Climate	Kansas Mesonet (K-State)	Portal CSV + API	5-60 min real-time	Open access	Low
Wildlife	eBird (Cornell Lab) - Bird Observations	eBird API + data products	Frequent	eBird terms; research use	Medium (sensitive species locations)
Wildlife	Kansas GAP Species Distribution Maps	Download GIS layers	Occasional updates	Open / state	Low
Demographics	US Census - Decennial Census + ACS	API + TIGER/Line + bulk	Annual (ACS); decennial	US Gov public domain	Low (aggregated)
Demographics	IPUMS NHGIS (National Historical GIS)	Download extracts	Occasional	Public domain underlying; attribution requested	Low
Agriculture	USDA NASS QuickStats	QuickStats API (JSON/CSV)	Annual + survey cycles	CC0	Low
Agriculture	USDA NASS Cropland Data Layer (CDL)	GeoTIFF download + WMS	Annual	Public domain/CC0	Low
Migration	IRS Migration Data (county-to-county)	Bulk files	Annual	Public domain	Low
Land	BLM General Land Office (GLO) Land Patents	Search + XML; bulk via indexing/scrape	Static historical	Public domain	Low/Medium (names)

Domain	Data source	Access	Cadence	License	Sensitivity
Land	PLSS Grid (Township-Range-Section)	Shapefile/Geodatabase (BLM/Kansas Geoportal)	Static	Public domain	Low
Land	Kansas GIS DASC Geoportal (Kansas Geoportal)	Download + REST Feature Services	Varies by layer	Open (gov data)	Low
Land	ORKA - Open Records for Kansas Appraisers	Web lookup; county GIS downloads vary	Near real-time	Public records / county terms	High (ownership/PII risk)
Hydrology	USGS WaterData / NWIS (Kansas)	REST API + WaterML/CSV/JSON	Hourly (streams) to periodic (gw)	Public domain	Low
Hydrology	KGS WIZARD - Water Levels / wells	Query UI + bulk CSV/ Shapefile/GeoJSON	Periodic/nightly updates	Public domain	Medium (private well locations)
Hydrology	USGS National Hydrography Dataset (NHD)	National Map download + services	Infrequent	Public domain	Low
Hydrology	Reservoir & River Conditions (USACE + Kansas Water Office)	Dashboards + CSV; scrape/API where available	Daily/hourly	Public domain	Low
Hydrology	Water Quality Portal (EPA/WQP) + KDHE monitoring	WQP download/API (CSV/JSON)	As sampled (monthly/quarterly)	Public domain	Low
Soils/Land Cover	NRCS SSURGO / gSSURGO	Soil Data Access API + downloads	Occasional updates	Public domain	Low
Soils/Land Cover	NLCD (National Land Cover Database)	GeoTIFF downloads	Periodic/annual products	Public domain	Low
Hazards	NOAA NCEI Storm Events Database	CSV downloads + query	Monthly	Public domain	Low
Hazards	NOAA SPC Severe Weather GIS	Shapefiles/KML/CSV	Annual	Open	Low
Hazards	FEMA Disaster Declarations (OpenFEMA)	Open data API (CSV/JSON)	Near real-time	Public domain	Low
Hazards	U.S. Drought Monitor	Shapefile/GeoJSON/ WMS	Weekly	Open	Low
Hazards	Wildfire Perimeters (NIFC + Kansas Forest Service)	Shapefiles + feature services	Ongoing/annual	Public domain	Low
Public Safety	FBI UCR/NIBRS Crime Data API	API	Annual/periodic	Public data	Medium (small-area reidentification)

Domain	Data source	Access	Cadence	License	Sensitivity
Public Health	Kansas Environmental Public Health Tracking (KDHE)	Portal endpoints	Varies	Gov open/public	High (health data)
Heritage	Kansas Historic Resources Inventory (KHRI)	Web search/map; GIS exports for public sites	Ongoing	Open with restrictions; sensitive sites generalized	High (site protection)
Heritage	National Register of Historic Places (NRHP) GIS	Shapefiles/feature service	As listed	Public domain	Medium
Heritage	Kansas Memory (Kansas Historical Society)	Web download + metadata	Continuous	Mostly public domain/free with attribution	Medium (personal info in documents)
Heritage	Chronicling America (Library of Congress) - Historic Newspapers	Open API + downloads	Static historical	Public domain (mostly up to 1922)	Low
Heritage	USGS Historical Topographic Map Collection	Download + WMS	Static archive	Public domain	Low
Geology	Kansas Geological Survey (KGS) data & maps	KGS portals + downloads	Varies	Open access (gov)	Medium (some well records)
Transportation	Kansas DOT GIS / ArcGIS Open Data	ArcGIS open data/feature services	Varies	Gov open	Low/Medium
Flood	Kansas Flood Mapping Dashboard + Floodplain layers	Web dashboard + hub datasets	Varies	Open	Low

8. Earth observation satellite data (open access) - ingestion strategy

The KFM Data Sources inventory includes a comprehensive list of open-access Earth observation satellites grouped by agency. KFM integrates these sources through a single pattern: treat satellite imagery products as STAC collections with Items by acquisition time and footprints, and store imagery as cloud-optimized GeoTIFFs or as externally-referenced assets when rehosting is not allowed.

8.1 Satellite ingestion pattern (STAC-first)

- One STAC Collection per product (e.g., Landsat Level-2 Surface Reflectance).
- Items per scene/tile/time; include sensors, bands, cloud cover, and processing level.
- Store only metadata + pointers when datasets are too large to mirror; cache derived tiles for map preview.
- Standardize CRS and band naming in derivatives; preserve original source references in PROV.

8.2 Satellite list referenced in KFM Data Sources

Satellite / mission (agency)
Terra (NASA)
Aqua (NASA)
Aura (NASA)
Landsat 8 & 9 (NASA/ USGS)
Suomi NPP (NASA/NOAA)
JPSS (NOAA-20, NOAA-21) (NOAA)
SMAP (NASA)
ICESat-2 (NASA)
OCO-2 (NASA)
NISAR (NASA/ISRO)
Sentinel-1A & 1B (ESA - Copernicus)
Sentinel-2A & 2B (ESA - Copernicus)
Sentinel-3A & 3B (ESA/EUMETSAT - Copernicus)
Sentinel-5P (ESA - Copernicus)
Sentinel-6 “Michael Freilich” (ESA/EuroCOM/ NASA)
ERS-1 & ERS-2 (ESA)
SMOS (ESA) (Soil Moisture Ocean Salinity)

Satellite / mission (agency)
CryoSat-2 (ESA)
Aeolus (ESA)
ISRO (India)
Resourcesat-1 & 2 (ISRO) (IRS-P6 & P7)
Oceansat-2 & 3 (ISRO)
INSAT-3D/3DR (ISRO/IMD) (METSAT geostationary)
NISAR (NASA/ ISRO)
ALOS “Daichi” (JAXA)
ALOS-2 “Daichi-2” (JAXA)
GCOM-C “Shikisai” (JAXA)
GOSAT (Ibuki) (JAXA/MOE)
Himawari-8 & 9 (JMA)
ADEOS-II “Midori-2” (JAXA)

9. Governance, FAIR/CARE, and sensitivity handling

KFM must treat some data as sensitive: private ownership, precise archaeological site locations, and certain health/public-safety indicators. Sensitivity is handled by: (a) policy labels at dataset/record/field level, (b) derivative datasets with explicit redaction provenance, and (c) fail-closed policy checks.

9.1 Sensitivity classes (recommended)

- **Public**: safe to publish without redaction.
- **Restricted**: requires role-based access (e.g., parcel ownership).
- **Sensitive-location**: coordinates must be generalized or suppressed (e.g., archaeology, sensitive species).
- **Aggregate-only**: only publish above thresholds (health/crime small counts).

9.2 Redaction as a first-class transformation

Redaction is a transformation that must be recorded in PROV. The raw dataset remains immutable; the redacted derivative is a separate DatasetVersion (often separate dataset_id) with a documented policy label.

9.3 CI policy regression suite

- Golden queries that previously leaked restricted fields must fail tests forever (non-regression).
- Negative tests: ensure sensitive-location layers cannot be returned at high precision to unauthorized roles.
- Field-level tests: verify owner names, health small counts, and exact archaeological coordinates are redacted.
- Audit integrity tests: every API response must include an audit reference and evidence bundle hash.

10. Operations, SLOs, and monitoring

Operational discipline prevents data sources from silently going stale. Each dataset carries a freshness SLO (e.g., Mesonet < 1 hour, Storm Events < 30 days, static archives exempt). Alerting triggers when SLOs are violated.

10.1 Observability signals

- Ingest runs: success/fail, duration, rows/bytes processed, retry counts.
- Freshness: last successful run timestamp per dataset + expected cadence.
- Quality drift: distribution checks, missingness, geometry errors.
- API: request latency, cache hit rate, policy denials, evidence resolution failures.
- Storage: object store growth, PostGIS index health, search index lag.

10.2 Backfill strategy

Backfills are explicit. The registry defines the historical coverage to backfill, the batching strategy, and expected runtimes. Backfills create new DatasetVersions with their own provenance; they never overwrite existing releases.

11. Implementation roadmap (build order that prevents rework)

To avoid the recurring failure mode of 'data sources get overlooked', the build must be driven by the registry: each dataset becomes a tracked integration deliverable with contract tests and promotion gates. The recommended build order is to ship governance and catalogs first, then add connectors in batches.

- **Milestone 0 - Skeleton**: trust membrane + policy scaffolding + audit logging; stub catalogs and a toy dataset.
- **Milestone 1 - Catalog core**: implement DCAT/STAC/PROV writers; enforce Raw->Work->Processed promotion in CI.
- **Milestone 2 - High-impact sources**: Census + NLCD/CDL + USGS WaterData + NOAA hazards (enables map/timeline).
- **Milestone 3 - Narrative/heritage**: Kansas Memory + Chronicling America + KHRI/NRHP with sensitivity controls.
- **Milestone 4 - Land/cadastral**: DASC base layers + PLSS + (restricted) ORKA adapter.
- **Milestone 5 - Domain expansion**: GBIF/iNaturalist/eBird + KGS geology + public health tracking.
- **Milestone 6 - Satellite STAC**: STAC proxy + caching for key products; derivative tile generation.

11.1 CI gates that must exist before adding more datasets

- Schema registry and validation tests for each connector.
- Golden-file tests for catalog outputs (DCAT/STAC/PROV).
- Policy regression tests (restricted fields must never leak).
- API contract tests for map/timeline and evidence resolution.
- Operational SLO checks (freshness for real-time sources; drift alerts).

Appendix A. Data source implementation profiles

Each profile below is intended to be directly convertible into an integration ticket: it defines connector behavior, validation gates, storage expectations, and policy considerations. These profiles are the primary mechanism to keep data sources from being forgotten during the build.

GBIF - Global Biodiversity Information Facility

Domain: Biodiversity **Primary data type:** Species occurrence points
Access: Open API + bulk downloads (DwC-A/CSV)
Update cadence: Continuous / as published **License:** Open data (varies by record; typically CC0/CC BY)
Sensitivity: Low **Source reference:** KFM Data Sources p.1

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Continuous / as published
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Observation location	Place (point)	Use WGS84; if obscured, store generalized geometry + sensitivity flag.
Observation time	Observation.time	Store eventDate and observedAt separately when available.
Taxon	Entity: Taxon	Store scientificName + taxonID; link to external taxonomy keys.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Prefer Darwin Core (DwC) semantics: occurrenceID, scientificName, eventDate, decimalLatitude/Longitude, basisOfRecord. Preserve original taxon keys for reconciliation.
- Implement incremental harvest by datasetKey + modified date when possible; otherwise maintain snapshot versions by publisher and deduplicate by occurrenceID.
- Respect per-record licenses (CC0/CC BY) and attribution; propagate to evidence citations.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.

- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

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iNaturalist Observations

Domain: Biodiversity **Primary data type:** Citizen-science observations (points + media)
Access: Open API + bulk exports
Update cadence: Continuous **License:** Varies (observation/media licenses)
Sensitivity: Medium (precise locations for sensitive species) **Source reference:** KFM Data Sources p.1

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Continuous
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Observation location	Place (point)	Use WGS84; if obscured, store generalized geometry + sensitivity flag.
Observation time	Observation.time	Store eventDate and observedAt separately when available.
Taxon	Entity: Taxon	Store scientificName + taxonID; link to external taxonomy keys.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Treat observation and media licenses separately; do not redistribute media unless license allows.
- Honor geoprivacy fields (obscured/private coordinates). Store true coordinate only if provided and policy permits; otherwise store generalized geometry.
- Use taxon IDs to align with GBIF taxonomies; keep original community taxon vs. observer taxon as separate claims.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.

- Backfill strategy documented (historical ranges and expected runtime).

NOAA NCEI - Historical Climate Data (Climate Data Online)

Domain: Climate **Primary data type:** Station time series (temperature, precipitation, etc.)
Access: API + bulk
Update cadence: Daily/monthly updates **License:** US Gov public domain
Sensitivity: Low **Source reference:** KFM Data Sources p.2

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Daily/monthly updates
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Station/site	SensorStation	Station metadata is slowly changing; treat as dimension table with versioning.
Measurement time series	Observation (timeseries)	Parameter codes + units required; keep qualifiers/detection limits.
Watershed joins (optional)	Place joins	Join station to admin units + NHD flowlines; store join confidence.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.

- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

Kansas Mesonet (K-State)

Domain: Climate **Primary data type:** High-frequency station observations
Access: Portal CSV + API
Update cadence: 5-60 min real-time **License:** Open access
Sensitivity: Low **Source reference:** KFM Data Sources p.2 / p.33

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	5-60 min real-time
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Station/site	SensorStation	Station metadata is slowly changing; treat as dimension table with versioning.
Measurement time series	Observation (timeseries)	Parameter codes + units required; keep qualifiers/detection limits.
Watershed joins (optional)	Place joins	Join station to admin units + NHD flowlines; store join confidence.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

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eBird (Cornell Lab) - Bird Observations

Domain: Wildlife **Primary data type:** Checklists (points + effort + taxa)
Access: eBird API + data products
Update cadence: Frequent **License:** eBird terms; research use
Sensitivity: Medium (sensitive species locations) **Source reference:** KFM Data Sources p.3

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Frequent
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed

Config key	Value / rule
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Observation location	Place (point)	Use WGS84; if obscured, store generalized geometry + sensitivity flag.
Observation time	Observation.time	Store eventDate and observedAt separately when available.
Taxon	Entity: Taxon	Store scientificName + taxonID; link to external taxonomy keys.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Model checklist effort (duration, distance, protocol) as first-class fields; it is critical for analysis and AI explanations.
- Implement sensitive-species policy gates; for restricted species, return generalized locations at query time.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

Kansas GAP Species Distribution Maps

Domain: Wildlife **Primary data type:** Species range polygons/rasters
Access: Download GIS layers
Update cadence: Occasional updates **License:** Open / state
Sensitivity: Low **Source reference:** KFM Data Sources p.3

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Occasional updates
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed

Config key	Value / rule
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Observation location	Place (point)	Use WGS84; if obscured, store generalized geometry + sensitivity flag.
Observation time	Observation.time	Store eventDate and observedAt separately when available.
Taxon	Entity: Taxon	Store scientificName + taxonID; link to external taxonomy keys.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.

- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

US Census - Decennial Census + ACS

Domain: Demographics **Primary data type:** Tabular + boundary geographies
Access: API + TIGER/Line + bulk
Update cadence: Annual (ACS); decennial **License:** US Gov public domain
Sensitivity: Low (aggregated) **Source reference:** KFM Data Sources p.4

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Annual (ACS); decennial
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Areal unit	Place (polygon, time-valid)	Key by FIPS/GEOID + vintage; boundaries change over time.
Statistic value	Observation.value	Store unit + denominator metadata; keep margins of error if present.
Flow (origin->dest)	Edge: MigrationFlow	Represent as directed edges with weights; aggregate thresholds enforced.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Store areal units as time-valid geometries (county/tract definitions change). Use a geometry version table keyed by vintage (e.g., TIGER year) and a crosswalk layer for joins.
- Support areal interpolation (optional) when comparing variables across boundary vintages; provenance must record the interpolation method and weights.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

IPUMS NHGIS (National Historical GIS)

Domain: Demographics **Primary data type:** Historic census tables + time-valid boundaries
Access: Download extracts
Update cadence: Occasional **License:** Public domain underlying; attribution requested
Sensitivity: Low **Source reference:** KFM Data Sources p.5

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Occasional
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Areal unit	Place (polygon, time-valid)	Key by FIPS/GEOID + vintage; boundaries change over time.
Statistic value	Observation.value	Store unit + denominator metadata; keep margins of error if present.
Flow (origin->dest)	Edge: MigrationFlow	Represent as directed edges with weights; aggregate thresholds enforced.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Store areal units as time-valid geometries (county/tract definitions change). Use a geometry version table keyed by vintage (e.g., TIGER year) and a crosswalk layer for joins.
- Support areal interpolation (optional) when comparing variables across boundary vintages; provenance must record the interpolation method and weights.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

USDA NASS QuickStats

Domain: Agriculture **Primary data type:** County/state ag statistics (crops, yields, livestock, economics)
Access: QuickStats API (JSON/CSV)
Update cadence: Annual + survey cycles **License:** CC0
Sensitivity: Low **Source reference:** KFM Data Sources p.6

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Annual + survey cycles
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
County/year stats	Observation	Use county GeoID + year; store commodity + unit metadata.
Raster pixels (CDL/NLCD)	RasterLayer (STAC)	One item per year; build tiles/overviews for UI.
Soil polygons	Feature (PostGIS)	SSURGO is relational; materialize common views for speed.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Keep NASS commodity, statistic category, and unit descriptors; these are required to avoid misinterpretation in Focus Mode narratives.
- Implement caching for common county-year-commodity queries to reduce API churn.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.

- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

USDA NASS Cropland Data Layer (CDL)

Domain: Agriculture **Primary data type:** Raster crop-type land cover (30m)
Access: GeoTIFF download + WMS
Update cadence: Annual **License:** Public domain/CC0
Sensitivity: Low **Source reference:** KFM Data Sources p.6

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Annual
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.

Source concept	Canonical concept	Implementation notes
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
County/year stats	Observation	Use county GeoID + year; store commodity + unit metadata.
Raster pixels (CDL/NLCD)	RasterLayer (STAC)	One item per year; build tiles/overviews for UI.
Soil polygons	Feature (PostGIS)	SSURGO is relational; materialize common views for speed.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Ingest as STAC: one Collection per product, one Item per year (or per quad for topo maps), with cloud-optimized GeoTIFFs where possible.
- Generate overview tiles + vector footprints for fast map preview; keep originals as authoritative assets.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.

- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

IRS Migration Data (county-to-county)

Domain: Migration **Primary data type:** Aggregate migration flows
Access: Bulk files
Update cadence: Annual **License:** Public domain
Sensitivity: Low **Source reference:** KFM Data Sources p.8-9

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Annual
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Areal unit	Place (polygon, time-valid)	Key by FIPS/GEOID + vintage; boundaries change over time.

Source concept	Canonical concept	Implementation notes
Statistic value	Observation.value	Store unit + denominator metadata; keep margins of error if present.
Flow (origin->dest)	Edge: MigrationFlow	Represent as directed edges with weights; aggregate thresholds enforced.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.

- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

BLM General Land Office (GLO) Land Patents

Domain: Land **Primary data type:** Historic patent records + scanned documents
Access: Search + XML; bulk via indexing/scrape
Update cadence: Static historical **License:** Public domain
Sensitivity: Low/Medium (names) **Source reference:** KFM Data Sources p.9

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Static historical
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Parcel / cadastral unit	Place/Parcel	Treat as restricted by default; derive non-identifying summaries.
Infrastructure feature	Feature	Normalize geometry + attributes; version as layers by publish date.
Legal land description	Text + PLSS join	Parse township-range-section; join to PLSS grid for mapping.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

PLSS Grid (Township-Range-Section)

Domain: Land **Primary data type:** Cadastral boundaries
Access: Shapefile/Geodatabase (BLM/Kansas Geoportal)
Update cadence: Static **License:** Public domain
Sensitivity: Low **Source reference:** KFM Data Sources p.9 / p.34

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Static
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Parcel / cadastral unit	Place/Parcel	Treat as restricted by default; derive non-identifying summaries.
Infrastructure feature	Feature	Normalize geometry + attributes; version as layers by publish date.
Legal land description	Text + PLSS join	Parse township-range-section; join to PLSS grid for mapping.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.

- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

Kansas GIS DASC Geoportal (Kansas Geoportal)

Domain: Land **Primary data type:** Authoritative base layers (boundaries, transportation, hydro, land cover, elevation, etc.)
Access: Download + REST Feature Services
Update cadence: Varies by layer **License:** Open (gov data)
Sensitivity: Low **Source reference:** KFM Data Sources p.34

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Varies by layer
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Parcel / cadastral unit	Place/Parcel	Treat as restricted by default; derive non-identifying summaries.
Infrastructure feature	Feature	Normalize geometry + attributes; version as layers by publish date.
Legal land description	Text + PLSS join	Parse township-range-section; join to PLSS grid for mapping.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

ORKA - Open Records for Kansas Appraisers

Domain: Land **Primary data type:** Parcel ownership + CAMA attributes (by county)
Access: Web lookup; county GIS downloads vary
Update cadence: Near real-time **License:** Public records / county terms
Sensitivity: High (ownership/PII risk) **Source reference:** KFM Data Sources p.34-35

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Near real-time
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries

Config key	Value / rule
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Parcel / cadastral unit	Place/Parcel	Treat as restricted by default; derive non-identifying summaries.
Infrastructure feature	Feature	Normalize geometry + attributes; version as layers by publish date.
Legal land description	Text + PLSS join	Parse township-range-section; join to PLSS grid for mapping.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Treat ownership and assessed values as **restricted** by default. Implement redaction layers (e.g., owner name removed) and role-based access via policy.
- Build a county-by-county adapter because parcel schemas vary. Capture per-county schema maps as governed configuration artifacts.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

USGS WaterData / NWIS (Kansas)

Domain: Hydrology **Primary data type:** Surface water + groundwater stations; time series
Access: REST API + WaterML/CSV/JSON
Update cadence: Hourly (streams) to periodic (gw) **License:** Public domain
Sensitivity: Low **Source reference:** KFM Data Sources p.33

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Hourly (streams) to periodic (gw)
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries

Config key	Value / rule
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Station/site	SensorStation	Station metadata is slowly changing; treat as dimension table with versioning.
Measurement time series	Observation (timeseries)	Parameter codes + units required; keep qualifiers/detection limits.
Watershed joins (optional)	Place joins	Join station to admin units + NHD flowlines; store join confidence.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Model each site as a SensorStation with geospatial point + period-of-record; observations are time-series with parameter codes.
- Implement incremental pulls using begin/end date windows and store raw WaterML where available to preserve qualifiers.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

KGS WIZARD - Water Levels / wells

Domain: Hydrology **Primary data type:** Well records + water levels
Access: Query UI + bulk CSV/Shapefile/GeoJSON
Update cadence: Periodic/nightly updates **License:** Public domain
Sensitivity: Medium (private well locations) **Source reference:** KFM Data Sources p.33

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Periodic/nightly updates
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries

Config key	Value / rule
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Station/site	SensorStation	Station metadata is slowly changing; treat as dimension table with versioning.
Measurement time series	Observation (timeseries)	Parameter codes + units required; keep qualifiers/detection limits.
Watershed joins (optional)	Place joins	Join station to admin units + NHD flowlines; store join confidence.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Well locations and depths can be sensitive in aggregate; enforce policy on fine-grained exports and avoid exposing private well owner info.
- Link wells to PLSS sections where possible to support historical land description joins.
- Geology involves multiple subdomains: stratigraphy, maps, wells, cores, production, industrial minerals. Model each as a separate Dataset under a shared KGS publisher.
- Some datasets have disclaimers and revision timing differences; retain KGS-issued versioning and publish KFM DatasetVersions without overwriting originals.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.

- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

USGS National Hydrography Dataset (NHD)

Domain: Hydrology **Primary data type:** Streams/rivers/lakes network vectors
Access: National Map download + services
Update cadence: Infrequent **License:** Public domain
Sensitivity: Low **Source reference:** KFM Data Sources p.33

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Infrequent

Config key	Value / rule
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Station/site	SensorStation	Station metadata is slowly changing; treat as dimension table with versioning.
Measurement time series	Observation (timeseries)	Parameter codes + units required; keep qualifiers/detection limits.
Watershed joins (optional)	Place joins	Join station to admin units + NHD flowlines; store join confidence.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).

- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

Reservoir & River Conditions (USACE + Kansas Water Office)

Domain: Hydrology **Primary data type:** Reservoir levels, releases, basin summaries
Access: Dashboards + CSV; scrape/API where available
Update cadence: Daily/hourly **License:** Public domain
Sensitivity: Low **Source reference:** KFM Data Sources p.33

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Daily/hourly
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Station/site	SensorStation	Station metadata is slowly changing; treat as dimension table with versioning.
Measurement time series	Observation (timeseries)	Parameter codes + units required; keep qualifiers/detection limits.
Watershed joins (optional)	Place joins	Join station to admin units + NHD flowlines; store join confidence.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.

- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

Water Quality Portal (EPA/WQP) + KDHE monitoring

Domain: Hydrology **Primary data type:** Water chemistry & sampling observations
Access: WQP download/API (CSV/JSON)
Update cadence: As sampled (monthly/quarterly) **License:** Public domain
Sensitivity: Low **Source reference:** KFM Data Sources p.33

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	As sampled (monthly/quarterly)
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Station/site	SensorStation	Station metadata is slowly changing; treat as dimension table with versioning.

Source concept	Canonical concept	Implementation notes
Measurement time series	Observation (timeseries)	Parameter codes + units required; keep qualifiers/detection limits.
Watershed joins (optional)	Place joins	Join station to admin units + NHD flowlines; store join confidence.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Treat sampling events as Observations with parameterCode, resultValue, unit, method, and QA qualifiers; keep detection limits and remark codes.
- Cache station metadata and spatial joins to NHD flowlines for watershed reasoning.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

NRCS SSURGO / gSSURGO

Domain: Soils/Land Cover **Primary data type:** Soil polygons + attributes; optional raster
Access: Soil Data Access API + downloads
Update cadence: Occasional updates **License:** Public domain
Sensitivity: Low **Source reference:** KFM Data Sources p.13

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Occasional updates
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
County/year stats	Observation	Use county GeoID + year; store commodity + unit metadata.
Raster pixels (CDL/NLCD)	RasterLayer (STAC)	One item per year; build tiles/overviews for UI.
Soil polygons	Feature (PostGIS)	SSURGO is relational; materialize common views for speed.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Normalize map unit keys and component tables; SSURGO is relational. Provide a pre-joined materialized view for common queries (dominant component, hydrologic group, texture).
- Expose Soil Data Access queries through a governed adapter to keep downstream reproducibility.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.

- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

NLCD (National Land Cover Database)

Domain: Soils/Land Cover **Primary data type:** Land cover rasters (30m time series)
Access: GeoTIFF downloads
Update cadence: Periodic/annual products **License:** Public domain
Sensitivity: Low **Source reference:** KFM Data Sources p.13

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Periodic/annual products
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
County/year stats	Observation	Use county GeoID + year; store commodity + unit metadata.
Raster pixels (CDL/NLCD)	RasterLayer (STAC)	One item per year; build tiles/overviews for UI.
Soil polygons	Feature (PostGIS)	SSURGO is relational; materialize common views for speed.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Ingest as STAC: one Collection per product, one Item per year (or per quad for topo maps), with cloud-optimized GeoTIFFs where possible.
- Generate overview tiles + vector footprints for fast map preview; keep originals as authoritative assets.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.

- Backfill strategy documented (historical ranges and expected runtime).

NOAA NCEI Storm Events Database

Domain: Hazards **Primary data type:** Severe weather event records
Access: CSV downloads + query
Update cadence: Monthly **License:** Public domain
Sensitivity: Low **Source reference:** KFM Data Sources p.31

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Monthly
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Event record	Event	Normalize event type taxonomy; keep original narrative description.
Event footprint	Geometry (point/line/polygon)	Tracks/perimeters as authoritative; points carry uncertainty.
Event time range	Event.time_range	Store start/end and reported timestamps separately; keep timezone or UTC.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Normalize event footprints and timestamps; store both reported location and derived spatial join results (county, watershed, ecoregion).
- For point reports (hail/wind) store uncertainty; for tracks/perimeters store polygons/lines as authoritative geometries.
- Maintain event-source crosswalks (Storm Events vs SPC) by date/time + spatial proximity; record match confidence.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.

- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

NOAA SPC Severe Weather GIS

Domain: Hazards **Primary data type:** Tornado tracks; hail/wind reports
Access: Shapefiles/KML/CSV
Update cadence: Annual **License:** Open
Sensitivity: Low **Source reference:** KFM Data Sources p.31

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Annual
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Event record	Event	Normalize event type taxonomy; keep original narrative description.
Event footprint	Geometry (point/line/polygon)	Tracks/perimeters as authoritative; points carry uncertainty.
Event time range	Event.time_range	Store start/end and reported timestamps separately; keep timezone or UTC.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Normalize event footprints and timestamps; store both reported location and derived spatial join results (county, watershed, ecoregion).
- For point reports (hail/wind) store uncertainty; for tracks/perimeters store polygons/lines as authoritative geometries.
- Maintain event-source crosswalks (Storm Events vs SPC) by date/time + spatial proximity; record match confidence.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.

- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

FEMA Disaster Declarations (OpenFEMA)

Domain: Hazards **Primary data type:** Disaster declarations + affected counties
Access: Open data API (CSV/JSON)
Update cadence: Near real-time **License:** Public domain
Sensitivity: Low **Source reference:** KFM Data Sources p.31

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Near real-time
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Event record	Event	Normalize event type taxonomy; keep original narrative description.
Event footprint	Geometry (point/line/polygon)	Tracks/perimeters as authoritative; points carry uncertainty.
Event time range	Event.time_range	Store start/end and reported timestamps separately; keep timezone or UTC.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Normalize event footprints and timestamps; store both reported location and derived spatial join results (county, watershed, ecoregion).
- For point reports (hail/wind) store uncertainty; for tracks/perimeters store polygons/lines as authoritative geometries.
- Maintain event-source crosswalks (Storm Events vs SPC) by date/time + spatial proximity; record match confidence.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.

- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

U.S. Drought Monitor

Domain: Hazards **Primary data type:** Weekly drought polygons (D0-D4)
Access: Shapefile/GeoJSON/WMS
Update cadence: Weekly **License:** Open
Sensitivity: Low **Source reference:** KFM Data Sources p.31

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Weekly
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Event record	Event	Normalize event type taxonomy; keep original narrative description.
Event footprint	Geometry (point/line/polygon)	Tracks/perimeters as authoritative; points carry uncertainty.
Event time range	Event.time_range	Store start/end and reported timestamps separately; keep timezone or UTC.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Normalize event footprints and timestamps; store both reported location and derived spatial join results (county, watershed, ecoregion).
- For point reports (hail/wind) store uncertainty; for tracks/perimeters store polygons/lines as authoritative geometries.
- Maintain event-source crosswalks (Storm Events vs SPC) by date/time + spatial proximity; record match confidence.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.

- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

Wildfire Perimeters (NIFC + Kansas Forest Service)

Domain: Hazards **Primary data type:** Fire perimeters polygons
Access: Shapefiles + feature services
Update cadence: Ongoing/annual **License:** Public domain
Sensitivity: Low **Source reference:** KFM Data Sources p.31

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Ongoing/annual
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Event record	Event	Normalize event type taxonomy; keep original narrative description.
Event footprint	Geometry (point/line/polygon)	Tracks/perimeters as authoritative; points carry uncertainty.
Event time range	Event.time_range	Store start/end and reported timestamps separately; keep timezone or UTC.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Normalize event footprints and timestamps; store both reported location and derived spatial join results (county, watershed, ecoregion).
- For point reports (hail/wind) store uncertainty; for tracks/perimeters store polygons/lines as authoritative geometries.
- Maintain event-source crosswalks (Storm Events vs SPC) by date/time + spatial proximity; record match confidence.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.

- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

FBI UCR/NIBRS Crime Data API

Domain: Public Safety **Primary data type:** Crime statistics
Access: API
Update cadence: Annual/periodic **License:** Public data
Sensitivity: Medium (small-area reidentification) **Source reference:** KFM Data Sources p.11

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Annual/periodic
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Source fields	Observation/Feature	Map via adapter-specific schema mapping file.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).

- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

Kansas Environmental Public Health Tracking (KDHE)

Domain: Public Health **Primary data type:** Health indicators dashboards + data extracts
Access: Portal endpoints
Update cadence: Varies **License:** Gov open/public
Sensitivity: High (health data) **Source reference:** KFM Data Sources p.43

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Varies
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Source fields	Observation/Feature	Map via adapter-specific schema mapping file.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Health metrics can be sensitive even when aggregated. Enforce minimum aggregation thresholds and suppress small counts per policy.
- Store indicator definitions and metadata alongside values to prevent misinterpretation.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

Kansas Historic Resources Inventory (KHRI)

Domain: Heritage **Primary data type:** Historic/archaeological site records
Access: Web search/map; GIS exports for public sites
Update cadence: Ongoing **License:** Open with restrictions; sensitive sites generalized
Sensitivity: High (site protection) **Source reference:** KFM Data Sources p.11

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Ongoing
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries

Config key	Value / rule
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Document/item	Artifact	Store persistent URL, media type, title, creator, date.
OCR text (derivative)	ArtifactText	OCR is a derived artifact; keep original scan as entity too.
Site record	Feature/Event	Sensitive archeological sites require generalized coordinates and access controls.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Exact coordinates for sensitive archaeological sites may be generalized and are legally protected; store sensitivity flags and enforce fail-closed access policies.
- Separate public site summaries from restricted location details; publish two derived datasets with separate provenance chains.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).

- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

National Register of Historic Places (NRHP) GIS

Domain: Heritage **Primary data type:** Historic places points/polygons
Access: Shapefiles/feature service
Update cadence: As listed **License:** Public domain
Sensitivity: Medium **Source reference:** KFM Data Sources p.11

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	As listed
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts

Config key	Value / rule
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Document/item	Artifact	Store persistent URL, media type, title, creator, date.
OCR text (derivative)	ArtifactText	OCR is a derived artifact; keep original scan as entity too.
Site record	Feature/Event	Sensitive archeological sites require generalized coordinates and access controls.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.

- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

Kansas Memory (Kansas Historical Society)

Domain: Heritage **Primary data type:** Digitized photos, letters, maps, documents
Access: Web download + metadata
Update cadence: Continuous **License:** Mostly public domain/free with attribution
Sensitivity: Medium (personal info in documents) **Source reference:** KFM Data Sources p.40

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Continuous
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.

Source concept	Canonical concept	Implementation notes
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Document/item	Artifact	Store persistent URL, media type, title, creator, date.
OCR text (derivative)	ArtifactText	OCR is a derived artifact; keep original scan as entity too.
Site record	Feature/Event	Sensitive archeological sites require generalized coordinates and access controls.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- OCR and historical texts are noisy; keep original scans and OCR outputs. Run text cleanup and entity extraction as derivative artifacts with explicit provenance.
- Index full-text for retrieval but require citations that point to the specific page/item/issue ID used.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.

- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

Chronicling America (Library of Congress) - Historic Newspapers

Domain: Heritage **Primary data type:** OCR newspaper pages + text
Access: Open API + downloads
Update cadence: Static historical **License:** Public domain (mostly up to 1922)
Sensitivity: Low **Source reference:** KFM Data Sources p.40

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Static historical
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.

Source concept	Canonical concept	Implementation notes
Document/item	Artifact	Store persistent URL, media type, title, creator, date.
OCR text (derivative)	ArtifactText	OCR is a derived artifact; keep original scan as entity too.
Site record	Feature/Event	Sensitive archeological sites require generalized coordinates and access controls.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- OCR and historical texts are noisy; keep original scans and OCR outputs. Run text cleanup and entity extraction as derivative artifacts with explicit provenance.
- Index full-text for retrieval but require citations that point to the specific page/item/issue ID used.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.

- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

USGS Historical Topographic Map Collection

Domain: Heritage **Primary data type:** Georeferenced topo map rasters (GeoTIFF)
Access: Download + WMS
Update cadence: Static archive **License:** Public domain
Sensitivity: Low **Source reference:** KFM Data Sources p.40

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Static archive
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Document/item	Artifact	Store persistent URL, media type, title, creator, date.

Source concept	Canonical concept	Implementation notes
OCR text (derivative)	ArtifactText	OCR is a derived artifact; keep original scan as entity too.
Site record	Feature/Event	Sensitive archeological sites require generalized coordinates and access controls.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Ingest as STAC: one Collection per product, one Item per year (or per quad for topo maps), with cloud-optimized GeoTIFFs where possible.
- Generate overview tiles + vector footprints for fast map preview; keep originals as authoritative assets.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

Kansas Geological Survey (KGS) data & maps

Domain: Geology **Primary data type:** Geologic maps, wells/cores, production data, industrial minerals
Access: KGS portals + downloads
Update cadence: Varies **License:** Open access (gov)
Sensitivity: Medium (some well records) **Source reference:** KFM Data Sources p.112+ / p.40

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Varies
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Well/core identifier	Feature	Normalize coordinates + depth; join to stratigraphic units where available.
Map publication	Artifact + Feature layers	Treat each map as artifact; GIS layers as derived assets with PROV.

Source concept	Canonical concept	Implementation notes
Production totals	Observation/Event	Store as time-series by county/field; keep reporting caveats.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Geology involves multiple subdomains: stratigraphy, maps, wells, cores, production, industrial minerals. Model each as a separate Dataset under a shared KGS publisher.
- Some datasets have disclaimers and revision timing differences; retain KGS-issued versioning and publish KFM DatasetVersions without overwriting originals.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

Kansas DOT GIS / ArcGIS Open Data

Domain: Transportation **Primary data type:** Roads, assets, plans (layer dependent)
Access: ArcGIS open data/feature services
Update cadence: Varies **License:** Gov open
Sensitivity: Low/Medium **Source reference:** KFM Data Sources p.43 / p.34

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Varies
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Parcel / cadastral unit	Place/Parcel	Treat as restricted by default; derive non-identifying summaries.
Infrastructure feature	Feature	Normalize geometry + attributes; version as layers by publish date.
Legal land description	Text + PLSS join	Parse township-range-section; join to PLSS grid for mapping.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.

- Backfill strategy documented (historical ranges and expected runtime).

Kansas Flood Mapping Dashboard + Floodplain layers

Domain: Flood **Primary data type:** Flood maps + floodplain polygons
Access: Web dashboard + hub datasets
Update cadence: Varies **License:** Open
Sensitivity: Low **Source reference:** KFM Data Sources p.43

Connector configuration (recommended defaults)

Config key	Value / rule
schedule	Varies
incremental_cursor	modified_date / eventDate / publicationDate when supported; otherwise snapshot+diff
auth	none (public) unless upstream requires keys; secrets stored in vault and never committed
rate_limit	respect provider limits; implement exponential backoff; cache common queries
format_targets	JSON/CSV for tabular; GeoJSON/Parquet for vectors; COG for rasters; PDF/JPEG/PNG for artifacts
policy_label	public | restricted | sensitive-location (set per source and sometimes per record)

Canonical mapping (minimum)

Source concept	Canonical concept	Implementation notes
Provider dataset identifier	Dataset / DatasetVersion	Persist as dataset_id + upstream_id; version is content-hash of raw manifest.
Source record identifier	source_record_id	Must be stable per upstream semantics; used in evidence citations.
Event record	Event	Normalize event type taxonomy; keep original narrative description.
Event footprint	Geometry (point/line/polygon)	Tracks/perimeters as authoritative; points carry uncertainty.
Event time range	Event.time_range	Store start/end and reported timestamps separately; keep timezone or UTC.

Integration design

KFM treats this source as a governed *Dataset* with explicit versioning. Each ingest run produces a **DatasetVersion** (checksums + run metadata), a **DCAT dataset record**, and a **PROV activity** linking raw assets to processed derivatives. Spatial assets (rasters/vectors) additionally register in a **STAC collection** to support map/timeline rendering and reproducible evidence citations.

Source-specific considerations

- Flood layers may come from multiple sources (research dashboards, FEMA FIRM derivatives, state hubs). Preserve source-of-truth identifiers and map scale/resolution metadata.
- Where datasets are published via ArcGIS Hub, use the REST endpoints to pull snapshots and record item IDs in provenance.

Ingestion workflow (raw -> work -> processed)

- **Discover**: resolve endpoints, parameters, and authentication needs; cache capability metadata.
- **Acquire**: fetch incremental slices when possible (by date/ID); otherwise snapshot and diff.
- **Normalize**: convert to canonical encodings (UTF-8), geometry (WGS84), and time (ISO 8601).
- **Validate**: schema checks, geometry validity, timestamp sanity, and license/policy checks.
- **Enrich**: derive join keys (GeoIDs), place/time normalization, and entity resolution candidates.
- **Publish**: promote to Processed, update catalogs (DCAT/STAC/PROV), and trigger index refresh (search/graph).

Minimum validation gates

- Row-level schema validation (required fields present; type coercion rules documented).
- Geometry validity + bounds (no self-intersections; inside expected extent when applicable).
- Temporal consistency (no future dates for historic archives; no negative durations).
- License + attribution captured in DCAT; restrictions encoded in policy.
- Provenance completeness: every promoted artifact has a PROV chain and a deterministic checksum.

Test plan (CI-ready)

- **Unit**: schema mapping and type coercion; geometry validity helpers; incremental window logic.
- **Integration**: run connector against a fixed small slice and assert stable checksums + counts.
- **Contract**: verify API response includes provenance bundle and respects policy redaction.
- **Regression**: dataset profiling metrics (null rates, min/max, distinct keys) are stable or explainably versioned.

Definition of Done (integration ticket)

- Connector implemented and registered in the data-source registry config.
- Raw acquisition produces deterministic manifest + checksums.
- Normalization emits canonical schema and/or STAC assets.
- Validation gates implemented and enforced in CI.
- Policy labels defined; restricted fields/locations are redacted per rules.
- Catalogs emitted (DCAT always; STAC/PROV as applicable) and link-check clean.
- API contract tests pass for at least one representative query.
- Backfill strategy documented (historical ranges and expected runtime).

Appendix B. Catalog templates (DCAT / STAC / PROV)

These templates define the minimum required fields. They should be implemented as schema-validated emitters in the catalog layer.

```
DCAT (minimum fields)
- dct:title, dct:description
- dct:publisher (org id)
- dct:license (SPDX or URL)
- dct:spatial (bbox or admin coverage)
- dct:temporal (start/end)
- dct:accrualPeriodicity (update cadence)
- dcat:distribution (download/API endpoints)
- prov:wasGeneratedBy (link to PROV activity)

STAC (minimum fields)
Collection:
- id, title, description, license
- extent.spatial (bbox), extent.temporal (interval)
- keywords, providers
Item:
- id, geometry, bbox, datetime
- properties: platform, instruments, proj:epsg, gsd (if known), eo:bands (optional)
- assets: { data, thumbnail, metadata } with roles + hrefs
- links: derived_from (PROV), self, collection

PROV (minimum fields)
Entity: raw_asset, normalized_table, derived_tile, ocr_text, etc.
Activity: ingest_run, transform_job, redaction_job
Agent: connector/service, data steward approval
Relations:
- wasGeneratedBy(entity, activity)
- used(activity, entity)
- wasDerivedFrom(derived, source)
- wasAssociatedWith(activity, agent)
```

Appendix C. Glossary

- **DCAT**: Data Catalog Vocabulary for dataset metadata.
- **STAC**: SpatioTemporal Asset Catalog for describing geospatial assets.
- **PROV**: W3C provenance model for lineage.
- **GeoID**: KFM internal identifier for a time-valid geographic unit.
- **Focus Mode**: The evidence-first narrative interface that must cite or abstain.
- **Promotion gate**: Automated + stewarded checks required to publish a dataset to Processed.